

Multimodal integration of metagenomic and clinical data for pathogen identification in lower respiratory tract infection

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Abstract

Distinguishing causative pathogens from incidental microbial detections remains a challenge in metagenomic diagnosis of respiratory infection. We developed AETIA (Agentic Evidence Tracing of Infectious Aetiology), which combines language models and prespecified rules to integrate metagenomic, clinical, microbiological and published evidence. In a retrospective development cohort of 55 patients with suspected lower respiratory tract infection at three hospitals, AETIA achieved 71.6% precision and 91.4% recall against physician-adjudicated pathogens. Its F1 score, summarizing precision and recall, was 80.3%, compared with 14.8% for unfiltered sequencing reports, 50.0% for multiplex polymerase chain reaction and galactomannan testing, and 45.5% for culture. In a 41-patient subset, AETIA exceeded four language models directly prompted with the same data by 13.6–15.1 F1 points. Each organism’s selection or exclusion was linked to the underlying evidence. These findings support integrating clinical context with metagenomic results for pathogen attribution; external validation and prospective evaluation are needed to establish clinical utility.

Introduction

Lower respiratory tract infection is frequently treated without a microbiological diagnosis. In a large prospective study of adults admitted with community-acquired pneumonia, no pathogen

was identified in 62% despite extensive conventional testing¹. Diagnostic yield is lower in critically ill patients, in whom previous antibiotic treatment reduces culture sensitivity and the differential diagnosis includes fungi, viruses and opportunistic pathogens. Metagenomic next-generation sequencing (mNGS) was developed to address this gap by detecting bacterial, fungal, viral and parasitic nucleic acids without a prespecified target^{2,3}. Clinical studies have shown its diagnostic potential in meningoencephalitis⁴, bloodstream infection^{5,6} and lower respiratory tract infection^{7–10}, and the assay is increasingly used in critically ill patients in Asia^{11,12}.

The main challenge is interpretation rather than detection. Respiratory mNGS identifies upper-airway flora^{13,14}, reagent and environmental contaminants¹⁵, reactivated herpesviruses and airway-colonizing *Candida*, as well as causative pathogens. In published series, this breadth has increased sensitivity at the cost of specificity. Across 511 clinical specimens, mNGS had a sensitivity of 50.7% and a specificity of 85.7% for infectious disease¹¹. In a prospective bronchoalveolar-lavage cohort, it identified a definite or probable pathogen in 65% of patients with lower respiratory tract infection, compared with 20% by culture, but had a specificity of 75.4% and a positive predictive value of 78.5% against a composite reference standard⁹. Distinguishing infection from colonization therefore remains a central limitation^{3,10}. Read thresholds and subtraction of no-template controls reduce background signal but do not resolve the biological ambiguity of a colonized respiratory specimen^{16,17}. Interpretation also requires findings outside the sequencing assay, including imaging, inflammatory trends, culture results and the timing and anatomical site of each test^{17–19}.

Conventional tests provide complementary information. Culture establishes organism viability and enables phenotypic susceptibility testing^{18,20}; multiplex PCR provides rapid, validated target-specific results²¹; galactomannan contributes an independent mycological criterion for invasive fungal disease²²; and serology can identify infections with little or no detectable airway nucleic acid. Imaging establishes the anatomical syndrome required for the diagnosis of pneumonia¹⁸. Specimen choice and timing impose further constraints: infection confined to blood, tissue, an abscess or an empyema may be absent from respiratory fluid, and intermittent or treated infection may be missed outside its detection window^{17,19,23}. Determining causation consequently requires integration of sequencing, conventional microbiology and clinical context.

Large language models could assist with this integration because they encode clinical knowledge²⁴, generate differential diagnoses from vignettes^{25,26} and have been evaluated as infectious-diseases consultants²⁷. Their performance on curated tasks, however, has not consistently transferred to clinical records. In an evaluation of 2,400 patient cases, current models performed worse than physicians, did not reliably follow diagnostic or treatment guidelines and had difficulty interpreting laboratory results²⁸. Model outputs also vary between runs and may not reveal which evidence determined a prediction²⁹. A generated explanation cannot by itself establish fidelity because a model’s stated reasoning may not correspond to the computation that produced its answer³⁰; in clinical applications, such post-hoc explanations can provide false reassurance³¹. Multi-agent systems^{32–34} and retrieval augmentation^{35,36} address specific components of this problem, but clinical use also requires reproducible decisions, traceable evidence, protection against reference-label leakage³⁷ and separation of prediction from explanation.

We developed AETIA (Agentic Evidence Tracing of Infectious Aetiology) to integrate res-

piratory mNGS with conventional microbiology and clinical data. Modality-specific language-model agents extract structured evidence; deterministic rules select candidate pathogens; and case-matched PubMed evidence can rescue candidates not selected initially. Explanations are generated only after the selected list has been fixed. We evaluated accuracy, organism burden, reproducibility and evidence traceability in patients with suspected lower respiratory tract infection from three medical centres in Taiwan and compared AETIA with individual diagnostic read-outs and direct prompting of general-purpose language models. The system is intended to provide an evidence-linked candidate list for physician review, not an autonomous diagnosis.

Results

AETIA separates evidence extraction, rule-based selection and literature-grounded rescue

AETIA processes nine clinical sources for each patient: admission diagnosis, underlying conditions, blood counts, other laboratory values, cultures, FilmArray, galactomannan, molecular microbiology and imaging, together with the laboratory mNGS report (Fig. 1a). A dedicated language-model agent converts each clinical source to a fixed schema. Each agent receives one modality and no reference labels. A deterministic summary combines the outputs into organism-specific evidence and host context.

Selection is rule-based (Fig. 1b). Each organism on the mNGS report is graded by its signal relative to negative controls, relative and absolute read burden, within-specimen dominance, anatomical alignment and support from hospital tests. These features and host immunosuppression determine a five-level causative ranking (Methods and Supplementary Tables 1 and 2). Prespecified guardrails demote common colonizers, oral flora, environmental organisms, skin flora, low-actionability herpesviruses, weak mould signals and non-specific laboratory labels. Organisms in the top two levels are selected; intermediate organisms require protected-pathogen status or direct support. A language-model safety review flags unselected organisms for further assessment but cannot alter or reorder the selected list.

For each selected or flagged candidate, the pipeline assembles literature, sequencing and clinical evidence (Fig. 1c). Up to ten PubMed articles are assessed first for species-specific evidence of infection at the target site and then for similarity to the patient’s specimen, syndrome, host, phenotype and timing. Separate modules quantify absolute and relative sequencing strength, exact-organism support from non-sequencing tests and colonization risk. Missing or pending tests remain unknown. The final rule retains every deterministic selection and adds a candidate only when the same organism is isolated from a sterile site or tissue, or when case-matched literature, site alignment and a qualifying sequencing signal are all present (Fig. 1d). Explanation generation begins only after this list has been fixed. The delivered table and traceable files report the evidence and rule associated with each selected and excluded organism (Fig. 4e).

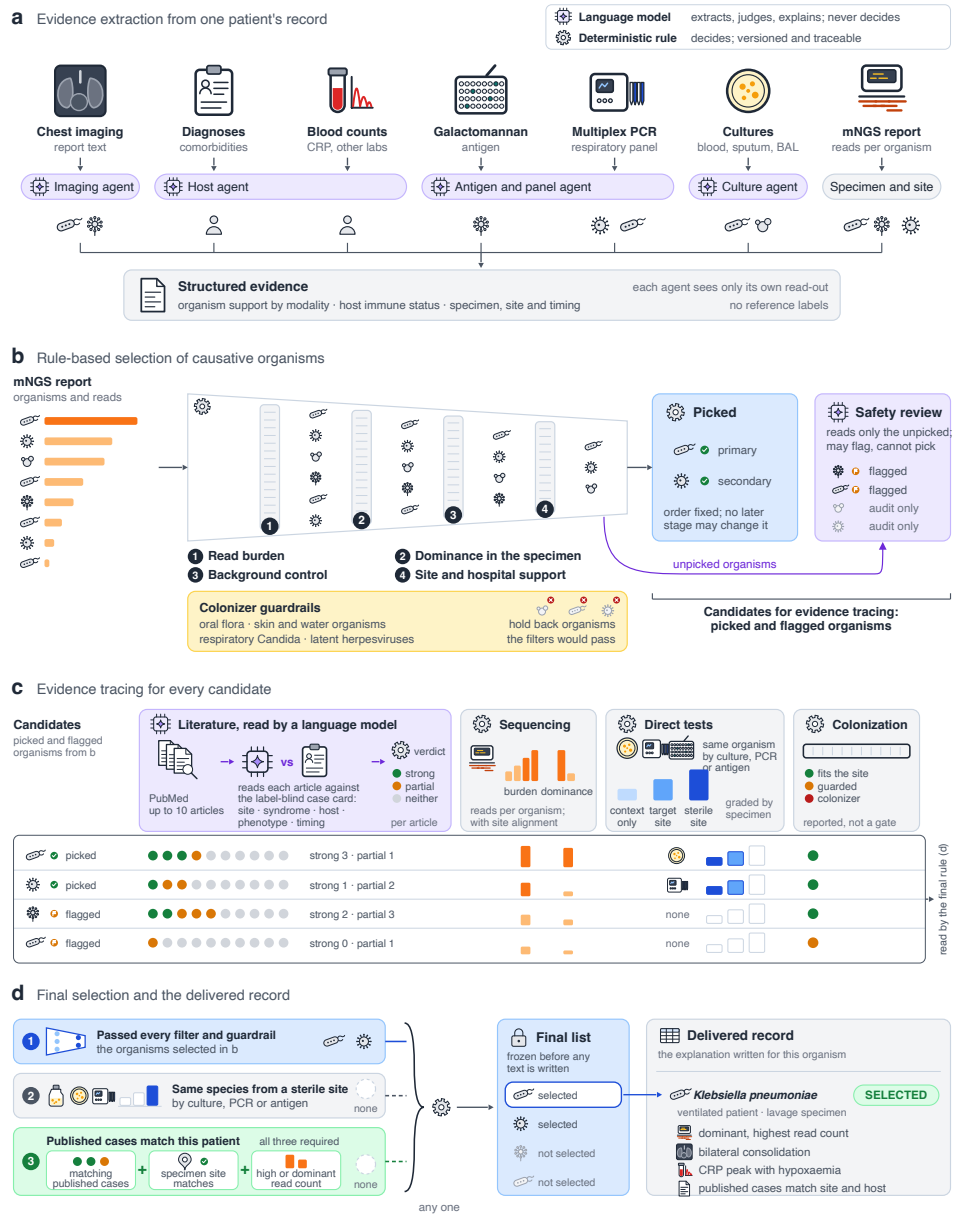


Figure 1: **Overview of AETIA.** Violet chip, language-model step (extracts, judges or explains; never decides); cog, deterministic rule (decides). **a**, Each read-out is parsed by a dedicated language-model agent into a fixed schema of organism-specific evidence, host status and specimen context; glyphs indicate the information available to each agent. Agents receive neither other read-outs nor reference labels. **b**, Every organism on the mNGS report (bars, reads) is assessed by four filters. Prespecified guardrails exclude colonizers and reactivated viruses (crossed glyphs). Selections are ordered once; the safety review may flag unselected organisms but cannot select them. **c**, Evidence matrix with one row per candidate. Up to ten PubMed articles are assessed against a label-blind case summary and classified by rule as strong, partial or neither (one dot per article). Additional columns show sequencing burden and dominance, exact-species direct tests by specimen and colonization status. **d**, A candidate is selected by any of three criteria (brace): passage through all filters and guardrails in **b**; isolation of the same species from a sterile site or tissue; or matching published cases together with anatomical agreement and a high or dominant read count. Dashed symbols indicate criteria not met by any candidate. The list is fixed before explanation generation. The delivered record reports each organism's disposition and the findings supporting it; explanatory text cannot add, remove or reorder organisms. Methods and Supplementary Tables 1–4. [d: prototype vignette (Fig. 4e); regenerate from the delivered record and approve.]

AETIA preserves pathogen recall while improving precision over individual tests

The primary evaluation included 55 patients with suspected lower respiratory tract infection from Kaohsiung Medical University Hospital (KMUH, $n = 30$), Taipei Veterans General Hospital (TVGH) and Tri-Service General Hospital (TSGH; combined $n = 25$), drawn from the 105-specimen registry (Table 1; Extended Data Fig. 1). At sampling, 71% were in intensive care, 56% had hospital- or ventilator-associated pneumonia, 37% had an active malignancy, 39% were receiving systemic corticosteroids and 60% had bronchoalveolar-lavage specimens. Treating infectious-diseases physicians provided the reference pathogen lists. Performance was evaluated at the exact-species (patient, organism) level using micro-averaged precision, recall and F1.

AETIA achieved 71.6% precision, 91.4% recall and 80.3% F1 (Fig. 2a,b). The unfiltered mNGS report had similar recall (93.1%) but 8.0% precision (F1 14.8%). FilmArray and galactomannan testing had 67.6% precision and 39.7% recall (F1 50.0%), whereas culture had 48.1% precision and 43.1% recall (F1 45.5%). Thus, AETIA retained most of the sensitivity of the unfiltered report while increasing precision approximately ninefold.

In a separate, earlier series of 35 KMUH intensive-care patients analysed with the first pipeline version, culture reported at least one organism in 24% of patients and FilmArray in 64%, whereas mNGS reported at least one organism in every patient (Fig. 2c,d). The mean number of organisms per patient was 5.76 for mNGS and 1.97 after pipeline processing. The overlap of this prototype series with the primary cohort remains to be established (Methods).

AETIA achieves higher recall than directly prompted general-purpose models

We compared AETIA with four general-purpose models (GPT-5.6 Sol, GPT-5.6 Luna, GPT-5.6 Terra and GPT-6 Astra) in 41 patients (KMUH $n = 30$, TVGH and TSGH $n = 11$). Each model received the same de-identified mNGS and clinical data as AETIA, one expert-written prompt and a structured output schema, without retrieval, a supplied candidate list or reference labels (Methods and Supplementary Note 2). Predictions were stored before evaluation.

The four directly prompted models had precision of 67.7–70.5%, recall of 56.6–60.5% and F1 of 62.4–63.9% (Fig. 3). In the same 41 patients, AETIA achieved 67.7% precision, 90.8% recall and 77.5% F1 (69 true positives, 33 false positives and 7 false negatives), exceeding the directly prompted models by 13.6–15.1 F1 points. The difference was attributable mainly to recall: GPT-6 Astra had the highest recall among the directly prompted models, but remained 30.3 percentage points below AETIA. [Slide 3 gives 71.62% precision here; the frozen tally TP 69 / FP 33 gives 67.65%. Resolve.]

AETIA records supporting evidence for 123 selections and reasons for 395 exclusions

The source records were multimodal and longitudinal. Among 32 KMUH admissions with available source workbooks, each admission contained a median of 247 recorded results (IQR 214–263; range 157–423) across laboratory testing, imaging and microbiology during a 31-day period, with new results on a median of 16 days and 1,452 words of free text (Fig. 4a–c). The records named

a median of eight organisms per admission (IQR 6–11). Of 270 organism–admission pairs, 81% were named by only one read-out and 91% were never isolated from a sterile site. Among 365 chest-imaging studies, 90% mentioned pneumonia, infiltrate or consolidation, but 44% of these reports qualified the finding (Fig. 4d). Culture reports with no growth were issued after a median of 6 days, compared with 3 days for reports with growth.

Across the 55-patient evaluation, the deterministic rules considered 518 candidates and selected 114; the final evidence rule added nine, yielding 123 selected candidates (2.2 per patient) and 395 exclusions (Fig. 4h). These nine additions could arise either from an exact-organism sterile-site or tissue isolate or from case-matched literature combined with site alignment and a qualifying sequencing signal. Each delivered record reports the selection or exclusion rule, supporting and opposing findings, specimen site and timing, sequencing burden and dominance, and any matched PubMed articles (Fig. 4e–g). A candidate absent from the mNGS report can enter through an exact-organism hospital test at a relevant site; an atypical pathogen supported only by IgM serology is retained for review. [Count the hospital-evidence selections among the 123 from the delivered records.]

Two prototype cases illustrate how the recorded evidence distinguishes respiratory colonization from invasive infection (Fig. 4e,f; Supplementary Note 3). In one ventilated patient, mNGS identified dominant *Klebsiella pneumoniae* (68,569 reads) and *Candida albicans*. New bilateral consolidation, pleural effusions, increasing C-reactive protein, hypoxaemia and a dominant sequencing signal supported *K. pneumoniae*; low, non-dominant reads, low-burden respiratory culture, absence of a sterile-site isolate and the respiratory-*Candida* guardrail supported exclusion of *C. albicans*. In a second patient, multiple blood cultures grew *Candida tropicalis* and bronchoalveolar reads were high, satisfying the direct-evidence criterion despite the guardrail. [Vignettes are from the January 2026 prototype; regenerate from the delivered pipeline outputs.]

For all cases, organism selection, the rule path and the structured evidence values were fixed before explanatory text was generated. The generator received the completed decision, and the audit required one entry for each selected organism, prohibited additional organisms and blocked incomplete output. This procedure establishes structural consistency with the completed decision; the clinical usefulness and record-level fidelity of the explanations were not independently evaluated.

Airway mNGS reports contain more organisms but have limited corroboration across tests

The three-hospital registry contained 105 mNGS specimens (Table 1; Extended Data Fig. 2). Airway specimens contained a median of five reported organisms (IQR 3–7; $n = 67$), compared with one (IQR 1–2; $n = 38$) in blood, cerebrospinal fluid, tissue and other sterile-site specimens. Apparent differences by illness severity disappeared after restriction to bronchoalveolar lavage: the median was five organisms in intensive care and on the ward, 5.5 with corticosteroid treatment and five without it, and five with and without acute respiratory distress syndrome. The difference between hospital- or ventilator-associated and community-acquired pneumonia persisted within lavage specimens (five versus three organisms). Thus, the number of organisms requiring interpretation was associated primarily with sampling an airway compartment.

Conventional tests corroborated a minority of reported organisms (Extended Data Fig. 2b). Of 414 organisms on the 105 mNGS reports, 75 (18.1%) had a same-species culture during the admission; conversely, 90 of 165 cultured organisms (54.5%) were absent from mNGS. Among 312 registry comparisons, 49.0% were sequencing detections without a culture counterpart, 29.8% were partial matches, 6.7% were mismatches and 5.8% were exact matches, using the registry's categories. Viruses accounted for 27.1% of reported organisms and 21.8% of adjudicated pathogens, and 100 of 112 reported viruses were herpesviruses.

Adjudicated pathogen lists were available in the shared workbook for a separate subset of 33 registry cases (Supplementary Note 6). Their mNGS reports contained 104 organisms, of which 38 (37%) were adjudicated as causative and 66 (63%) as colonizing, contaminating or incidental. Read count ranked a causative organism above a non-causative organism in only 61% of within-case pairs, and the distributions overlapped across five orders of magnitude. Ten species were causative in one patient and non-causative in another. For example, *Acinetobacter nosocomialis* with 189,467 reads in lavage fluid was adjudicated as colonization, whereas *Pneumocystis jirovecii* with 7 reads was adjudicated as causative.

The same 33-case analysis contained 55 adjudicated pathogens, 17 (31%) of which were absent from the mNGS report of the sequenced specimen. Another read-out identified 16 of these 17 pathogens: culture in eight, PCR in five, galactomannan in two and IgM serology in one. Fourteen were identified within two days of mNGS sampling, and 11 were identified in a different specimen type. Sequencing and conventional tests therefore had complementary failure modes.

The registry also contained 22 specimens (21.0%) without a recorded causative pathogen. Six had an explicit negative entry, whereas 16 had the dash used by one hospital for unlabelled rows. mNGS reported at least one organism in 18 of these 22 specimens, with 47 organisms in total and as many as eight in one specimen. [Confirm with the TVGH investigators whether the dash in the adjudication column means "no causative pathogen identified" or "not recorded"; if the former, these 22 specimens are the no-infection stratum the evaluation cohorts lack, and the baselines and AETIA should be scored on them.]

Nearly half of patients had treatment changes attributed to mNGS without a detectable survival difference

Antimicrobial treatment changed in 69 of the 105 registry patients (65.7%), and clinicians attributed a change to mNGS in 52 (49.5% of the registry; 58.4% of the 89 patients with recorded attribution). These changes comprised 36 escalations and 13 de-escalations; 13 of all 19 de-escalations were attributed to mNGS. The cohort was severely ill: 82.7% underwent mechanical ventilation within three days of sampling and 48.6% died in hospital. Reports associated and not associated with an attributed treatment change contained the same median number of organisms (four).

No survival difference was detected between the 52 patients whose treatment change was attributed to mNGS and the 37 whose change was not. In-hospital death occurred in 30 of 52 patients (57.7%) and 21 of 37 patients (56.8%), respectively (odds ratio 1.04, 95% confidence interval 0.45–2.41; $P = 1.00$, Fisher's exact test). Adjustment for intensive care at sampling,

acute respiratory distress syndrome, active malignancy, corticosteroid treatment and age gave a similar estimate (adjusted odds ratio 1.22, 95% confidence interval 0.46–3.21; $P = 0.69$). This retrospective attribution was not an assigned exposure, every patient underwent mNGS and the registry contained neither antimicrobial agents nor administration dates; the analysis therefore does not estimate a causal effect of mNGS-guided treatment or the appropriateness of therapy.

Early defervescence and improved oxygenation also occur in patients who subsequently die

We examined physiological trajectories in 48 KMH admissions with machine-readable complete discharge records (31 deaths and 17 survivors), using every dated laboratory value and vital sign and anchoring observations to the day of the mNGS specimen (Methods) (Extended Data Fig. 3). The analysis was descriptive because the records did not contain antimicrobial start dates, stop dates or coverage of the adjudicated pathogen.

Among the 31 patients who died, death occurred a median of 15 days after the mNGS specimen (IQR 7–21; range 1–57) (Extended Data Fig. 3a); 22 died more than 7 days and 17 more than 14 days after sampling. Every patient who was febrile in the two weeks before sampling—10 decedents and 5 survivors—was afebrile on days 3–7. Among decedents, the proportion with a maximum temperature of at least 38 °C fell from 39% before sampling to 18% on days 0–2 and 3% on days 3–7. Temperature fell by a median of 0.5 °C between the pre-sampling and day 3–7 windows ($n = 27$ paired; $P = 0.001$, Wilcoxon signed-rank). Arterial P_{O_2} increased by days 3–7 in 17 of 27 decedents. These changes were often transient: fever returned in 9 of 21 decedents whose temperature improved, and P_{O_2} subsequently fell in 12 of the 17 whose oxygenation improved.

Platelet trajectories differed descriptively between survivors and decedents (Extended Data Fig. 3c). Median platelet count in survivors increased from 169 before sampling to 257 on days 3–7 and 360 on days 8–14, whereas the corresponding values in decedents were 82, 48 and $55 \times 10^3 \mu\text{L}^{-1}$. Among 14 assessable decedents, the count declined by a median of $19 \times 10^3 \mu\text{L}^{-1}$ from the pre-sampling to day 3–7 window ($P = 0.004$) and improved in 2. Lactate, white-cell count, creatinine and procalcitonin did not show corresponding differences between the groups. These trajectories show that early defervescence and improved oxygenation were not confined to survivors; without treatment timing or coverage, they cannot establish whether the causative organism was correctly identified or treated.

Discussion

Respiratory mNGS findings required organism-level clinical interpretation. In the registry, 63% of reported organisms were not adjudicated pathogens, read count ranked a pathogen above a non-pathogen in only 61% of pairs and ten species were causative in some patients but not in others. Clinicians nevertheless attributed an antimicrobial change to mNGS in 49.5% of the registry, including 13 of 19 de-escalations. The patient’s subsequent course did not indicate whether the organism treated was the right one: every KMH patient febrile before the specimen was afebrile by day 3–7 and oxygenation improved in 17 of 27 patients who nevertheless died, so

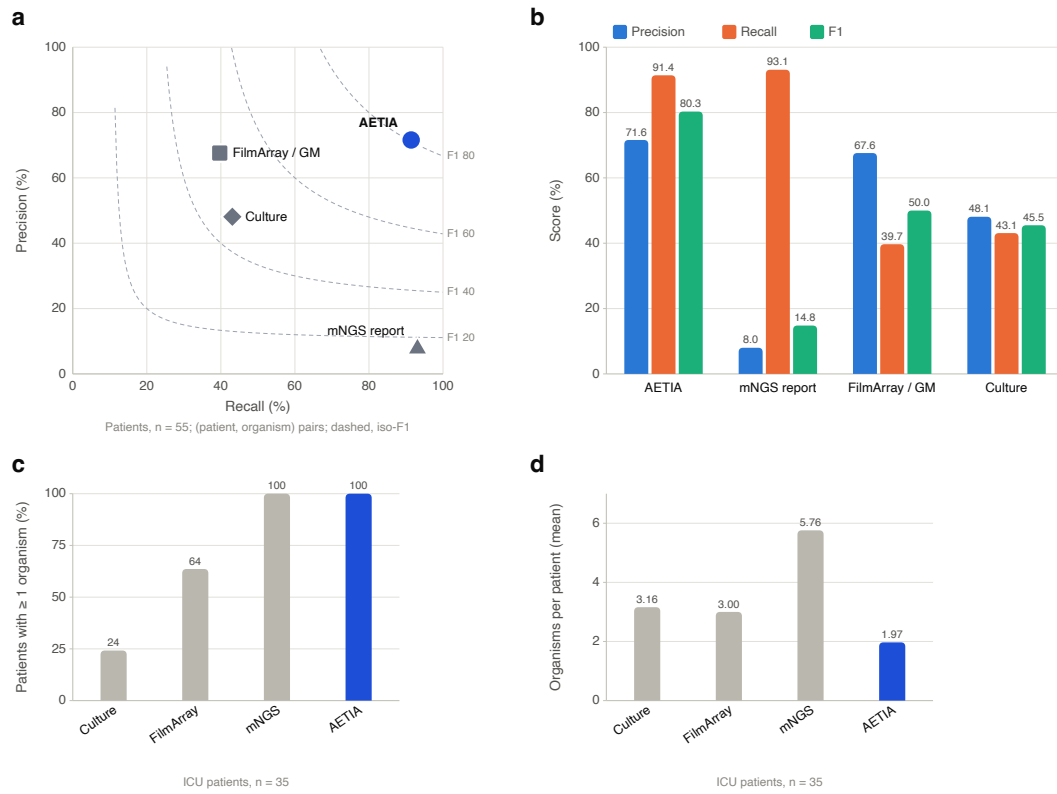


Figure 2: Pathogen selection compared with the raw mNGS report, FilmArray/galactomannan and culture. **a**, Precision against recall for AETIA (blue) and for each read-out taken at face value: all organisms in the mNGS report (triangle), all positive FilmArray or galactomannan targets (square) and all cultured organisms (diamond); dashed curves, constant F1. **b**, The same comparison as precision, recall and F1. $n = 55$ patients (KMUH 30; TVGH and TSGH 25); unit, (patient, organism) pair at species level; reference standard, physician-adjudicated pathogens (Methods). **c,d**, Earlier intensive-care series (KMUH, $n = 35$; first pipeline version): fraction of patients with at least one reported organism (**c**) and mean organisms per patient (**d**).

Table 1: **Characteristics of the three-hospital mNGS registry.** The 105-specimen registry from which the 55- and 41-patient evaluation cohorts were drawn (Extended Data Fig. 1). Values are n (%) or median (interquartile range).

Characteristic	All patients ($n = 105$)
<i>Hospital</i>	
KMUH	51 (48.6%)
TVGH	35 (33.3%)
TSGH	19 (18.1%)
Age, years	69 (56–75)
Male sex	68 (64.8%)
Current or former smoker	19 (18.1%)
Intensive care unit at sampling	75 (71.4%)
<i>Underlying conditions</i>	
Diabetes mellitus	29 (27.6%)
Chronic kidney disease	16 (15.2%)
End-stage renal disease on dialysis	4 (3.8%)
Liver cirrhosis	1 (1.0%)
Autoimmune disease	6 (5.7%)
Active malignancy	39 (37.1%)
Systemic corticosteroids [†]	40 (38.5%)
ARDS on the day of sampling [†]	48 (46.6%)
<i>Pneumonia category</i>	
Community-acquired	21 (20.0%)
Hospital- or ventilator-associated	59 (56.2%)
Not classified	25 (23.8%)
PaO ₂ /FiO ₂ ratio at sampling	200 (152–302)
<i>mNGS specimen</i>	
BALF	63 (60.0%)
Blood	24 (22.9%)
Sputum	4 (3.8%)
CSF	6 (5.7%)
Tissue	5 (4.8%)
Other	3 (2.9%)
Organisms per mNGS report, median (IQR)	4 (1–6)
mNGS report with no organism	8 (7.6%)
Any positive culture	69 (65.7%)
<i>Clinical course</i>	
Antimicrobial therapy changed	69 (65.7%)
Escalation	49 (46.7%)
De-escalation	19 (18.1%)
Change attributed to mNGS [‡]	52 (49.5%)
Mechanical ventilation within 3 days of sampling [†]	81 (82.7%)
In-hospital death [†]	51 (48.6%)

[†]Denominator excludes patients without a recorded value. [‡]Denominator is all patients; attribution was not recorded for 16 patients. KMUH, Kaohsiung Medical University Hospital; TVGH, Taipei Veterans General Hospital; TSGH, Tri-Service General Hospital; BALF, bronchoalveolar lavage fluid; CSF, cerebrospinal fluid.

the bedside signals read as confirmation of an empirical choice occurred in survivors and decedents alike and reversed in roughly half. In the 55-patient evaluation cohort, AETIA combined modality-specific extraction, deterministic scoring, safety review and case-matched literature to achieve 91.4% recall and 71.6% precision. It retained the sensitivity of the unfiltered mNGS report while substantially improving precision and outperformed individual conventional read-outs. On 41 patients, it also exceeded four general-purpose language models prompted directly with the same data by 13.6–15.1 F1 points. These findings support structured integration of sequencing, conventional microbiology and clinical data rather than interpretation by a single assay threshold or model prompt.

The comparison with direct prompting suggests that the main advantage was task structure rather than model capacity. The direct prompts specified colonization, anatomical-site and negative-control considerations, and their precision was similar to that of AETIA. Their lower recall indicates that general caution was applied too broadly. A deterministic rule can encode organism-specific exceptions, such as lower read thresholds for protected pathogens or additional requirements for environmental organisms. The literature stage then evaluates whether published cases match the species, site, syndrome and host. [Replace the proposed *Stenotrophomonas* and *Pneumocystis* examples with verified cases from the per-patient comparison.] The small recall difference among the four directly prompted models further suggests that increasing model capacity alone may not resolve this formulation problem.

Traceability is a separate contribution. The clinical records contained many discordant and asynchronous results: a median of 247 recorded results and eight named organisms per admission, with only 19% of organism–admission pairs supported by more than one read-out. AETIA records the evidence supporting or opposing every selected organism and the reason each non-selected organism was excluded. Because the structured decision is fixed before an explanation is generated, the explanation can be checked against the rule path and source values. This differs from a model-generated account produced in the same pass as the prediction, which may not reflect the computation that determined the answer³⁰ and can provide false reassurance in clinical use³¹. This design does not establish that the explanations are clinically adequate, but it makes discrepancies identifiable and testable.

The study has several limitations. It was retrospective, included 55 patients in the primary analysis and 41 in the model comparison, and used data from three hospitals in one country and one sequencing laboratory. The reference standard was physician adjudication informed by the mNGS report and was therefore not independent of the index test. The rule set was developed with access to these labels, no patient was held out for external validation and the reported performance is a development-set estimate. Both evaluation cohorts required at least one adjudicated pathogen at a pulmonary site, so patients whose final diagnosis was infection elsewhere or no infection were excluded. This is a selection on the outcome: it defines a population that cannot be identified when the test is ordered, and it removes the patients in whom the site and colonization guardrails would be most consequential, since the correct output for them is an empty pulmonary list. The precision reported here therefore applies to patients in whom pneumonia was ultimately confirmed and should not be read as the false-positive rate a clinician would encounter when sending mNGS on suspicion. This missing stratum is clinically relevant:

22 of 105 registry specimens had no recorded causative pathogen, although the interpretation of 16 unlabelled entries requires confirmation. Scoring AETIA and the comparators in these patients is an important next analysis.

The pipeline also shares constraints with the underlying evidence. An organism detected only in a non-sterile specimen cannot be proven causative without corroborating clinical or microbiological findings. Guardrails that correctly remove most airway colonizers may also demote uncommon true infections, including *Candida* pneumonia, herpes simplex virus tracheobronchitis or galactomannan-negative aspergillosis. Conversely, 31% of adjudicated pathogens were absent from the mNGS report of the sequenced specimen; the pipeline can recover such pathogens only when another test identifies them. These errors require an additional specimen or assay, not greater sequencing depth. The literature and case-fit stages use a commercial language model and a changing PubMed corpus. Frozen article snapshots permit replay of the reported analysis, but future runs may retrieve different evidence and model judgements are not deterministic. The two illustrative cases and the organism-burden comparison originated from an earlier 35-patient series and require confirmation using the delivered records from the primary cohort. Finally, we assessed agreement with adjudication rather than clinical benefit. We did not determine whether the selected lists changed antimicrobial treatment or outcomes, nor did independent physicians assess the completeness, fidelity or usefulness of the explanations. The registry does record what the clinicians did next, and acting on the sequencing report was not associated with survival in it; that null is not evidence against benefit, because every registry patient was sequenced, the exposure is the treating team’s own retrospective attribution rather than an assignment, and no antimicrobial agent or date is recorded from which the appropriateness of a change or the time to appropriate therapy could be computed. The higher mortality among patients whose treatment changed measures the indication for changing it rather than its effect, and any endpoint measured in days is subject to immortal time, because a change credited to a specimen taken days into an admission can only be recorded in patients who survive to receive it. The trajectory analysis sharpens this gap without closing it. It establishes that the window is long—death followed the specimen by a median of 15 days—and that the response signals available at the bedside do not discriminate, but it is observational, every patient was sequenced, and antipyretics, corticosteroids, sedation and ventilator settings act directly on temperature and gas exchange and were not recorded in a form permitting adjustment. The temperature column of the whole-record extracts also required repair before use (Methods).

A prospective study should freeze the rules and prompts before enrolment, include patients without infection, use an adjudication committee blinded to the pipeline output and evaluate changes in antimicrobial treatment and clinical outcomes. Two analyses of the existing cohort should precede it. The KMH discharge records contain the antimicrobial charts, so the interval from the report to a change of therapy and the adequacy of that change against the adjudicated pathogen are extractable without enrolling anyone; and the 22 specimens without a recorded pathogen provide the no-infection stratum once their adjudication convention is confirmed. Explanation quality should be assessed independently by comparing decisions made with and without the evidence record and by auditing exclusion reasons in false-negative cases. The same architecture could incorporate resistance-gene evidence and be evaluated in bloodstream

and central nervous system infections. Such extensions should follow external validation of the present pathogen-selection rules.

Methods

Ethics and study oversight

This retrospective study analysed existing clinical records and mNGS reports. Patients were not contacted, and pipeline outputs were not used in clinical care. The study was approved by the institutional review boards of Kaohsiung Medical University Hospital (KMUH; [protocol number]), Taipei Veterans General Hospital (TVGH; [protocol number]) and Tri-Service General Hospital (TSGH; [protocol number]); each board waived the requirement for informed consent because the study used retrospectively collected, de-identified data.

Direct identifiers were removed before records were submitted to a commercial language-model service. Submissions were made through the provider’s application programming interface under terms that exclude the use of submitted data for model training, and reference labels were withheld from all model inputs. AETIA was evaluated as research software and returned candidate pathogens for physician review rather than a clinical diagnosis.

Study design and participants

We conducted a retrospective, multicentre diagnostic-accuracy study at three tertiary referral centres in Taiwan. Participants underwent mNGS for suspected lower respiratory tract infection at KMUH, TVGH or TSGH between August 2023 and September 2025. The index test was AETIA, and the reference standard was the causative-pathogen list adjudicated by the treating physicians. Comparators were conventional microbiological read-outs interpreted at face value and four general-purpose language models prompted directly with the same patient data.

The three hospitals maintained a shared registry of 105 mNGS specimens (KMUH, 51; TVGH, 35; TSGH, 19), together with demographics, comorbidities, conventional microbiology, mNGS reports, treatment changes and outcomes (Table 1). Two evaluation cohorts were defined before model inference (Extended Data Fig. 1). The primary cohort included 55 patients (KMUH, 30; TVGH and TSGH, 25) with a complete standardized record, a per-specimen mNGS report and at least one adjudicated causative pathogen at a pulmonary site. Patients whose adjudicated final diagnosis was infection at another site, such as urinary tract infection, or no infection, were therefore not evaluated, even where mNGS had been sent for suspected pneumonia. This restricts the estimates reported here to patients in whom lower respiratory tract infection was ultimately confirmed, a population that cannot be identified at the time the test is ordered; the consequences for the interpretation of precision are considered in the Discussion. [Give the number excluded on this ground and the site of infection in each, and state whether the exclusion was applied to the registry or to the delivered set.] The direct-prompting cohort included the 30 KMUH patients and the 11 TVGH and TSGH patients whose records were complete in every section ($n = 41$). [Reasons for the two-hospital exclusions (54 registry candidates, 48 with complete records, 25 in the primary cohort).] An earlier series of 35 KMUH intensive-care patients, analysed with the first pipeline version, was used only for the organism-burden

comparison (Fig. 2c,d) [state its overlap with the primary cohort].

Metagenomic sequencing and conventional microbiology

mNGS was ordered by the treating physicians as part of clinical care and performed by one clinical laboratory (Asia Pathogenomics, New Taipei City, Taiwan) using a previously described workflow^{38,39}. Specimens were predominantly bronchoalveolar lavage fluid (60%) and blood (23%; Table 1). Briefly, specimens were heat-inactivated and mechanically lysed. Nucleic acids were extracted without host depletion using the TIANMicrobe Pathogen DNA Kit (Tiangen) and, for RNA, the QIAamp Viral RNA Mini Kit (Qiagen)³⁸. Libraries were prepared with unique dual-index adapters (TIANSeq Direct Fast DNA Library Prep Kit, Tiangen) and sequenced as 75-bp single-end reads on an Element Biosciences AVITI system, yielding approximately 40 million reads per sample³⁹.

Reads were quality-filtered with Trimmomatic⁴⁰ and PRINSEQ⁴¹. Human reads were removed by alignment to T2T-CHM13⁴² with BWA-MEM⁴³, and the remaining reads were aligned to a curated database containing 5,884 viral, 18,432 bacterial, 3,146 fungal and 370 parasitic genomes³⁹. Extraction, library and no-template negative controls were included in each run. Organisms were reported when normalized abundance exceeded ten times the highest value in a concurrent negative control. For fastidious, low-biomass pathogens such as *Mycobacterium tuberculosis*, uniquely mapping reads were required in the specimen and absent from all controls³⁹. AETIA used only the resulting clinical report—the reported organisms, read counts and ranks in the negative-control comparison—and did not process raw sequencing reads.

Conventional microbiology was performed in each hospital’s accredited clinical laboratory as part of routine care: culture with identification by mass spectrometry or automated biochemical systems; the BioFire FilmArray Pneumonia Panel (bioMérieux) on lower respiratory specimens, which reports typical bacteria semi-quantitatively and atypical bacteria, viruses and resistance genes qualitatively^{21,38}; the Platelia *Aspergillus* galactomannan enzyme immunoassay (Bio-Rad) in serum and lavage fluid⁴⁴; and targeted molecular tests. Results were taken as reported. Within the pipeline, a FilmArray bacterial target contributes evidence in proportion to its semi-quantitative bin, and galactomannan is considered positive at an index of ≥ 1.0 in lavage fluid and ≥ 0.5 in serum, in line with consensus definitions²².

Reference standard

The reference standard was established by a clinical review committee, following the approach used in clinical mNGS studies in which physicians adjudicate the clinical significance of sequencing findings^{4,16}. The committee comprised three infectious-diseases physicians from the participating medical centres. For each patient, the committee reviewed the presentation, imaging, blood counts, inflammatory markers, conventional microbiology, mNGS report, clinical course and response to treatment. It classified each organism detected by any method as a causative pathogen of the current episode or as a colonizer, contaminant, reactivated virus or bystander. Clinical context was used to adjudicate organisms that culture cannot confirm, including viruses, fungi and fastidious bacteria. The resulting causative pathogen list (one to eight organisms per patient) was recorded in the registry. Two label revisions affecting four patients

were finalized before evaluation, and the final labels were used for all reported analyses. Labels were available only to the evaluation scripts and were excluded from pipeline inputs, prompts and configuration. Because the committee reviewed the mNGS report, the reference standard was not independent of the index test. The absence of independent adjudication and blinding during rule development is addressed in the Discussion.

Data extraction and standardization

Source records comprised per-patient workbooks at KMUH and a shared master workbook with case-presentation slides at TVGH and TSGH. Rule-based parsers divided workbook contents into clinical sections and extracted slide text; information contained only in images was transcribed from rendered slides. Slides were linked to registry cases using patient, date and file metadata.

A language model (GPT-5.6 Luna, OpenAI) then converted each raw section to a fixed schema for nine clinical domains: admission diagnosis, underlying conditions, complete blood count, other laboratory values, cultures, FilmArray, galactomannan, molecular microbiology and imaging. The mNGS report was retained as a tenth input. The same prompt and output schema were used for all records. Clinical observations were restricted to a window anchored on collection of the mNGS specimen, and records without a parsable time were excluded. Hospital-specific patient identifiers and specimen codes were used to link records while preventing collisions between institutions.

Overview of AETIA

AETIA analyses one patient at a time in five stages (Fig. 1). First, modality-specific language models convert the standardized record into structured clinical evidence. Second, prespecified rules combine the clinical evidence with the mNGS signal to make an initial pathogen selection. Third, a safety review identifies unselected organisms that warrant additional evaluation but cannot change the initial selection. Fourth, the system evaluates the retained candidates against published cases, sequencing strength, direct microbiological evidence and anatomical plausibility, after which a rule-based integration step fixes the final list. Finally, a language model explains the fixed decision.

Language models were therefore used to extract or assess evidence, whereas version-controlled rules made every selection. Each model task used a fixed prompt and structured output schema (Supplementary Table 5). The pipeline recorded the rule applied at each decision and cryptographic hashes of the corresponding inputs. Its output comprised the selected pathogen list and, for every organism considered, the evidence for and against selection and the rule that determined its disposition. Explanation generation occurred only after selection and could not alter the list.

Modality-specific evidence extraction

Five language-model agents (GPT-5, OpenAI) processed distinct parts of the standardized record: host factors and laboratory measurements; culture; FilmArray and galactomannan; molecular microbiology; and imaging. Each agent received only its assigned clinical domain,

used a fixed prompt and output schema and had no access to the other agents or the reference labels. The host agent summarized immunosuppression, malignancy, corticosteroid exposure, renal failure and related factors as prespecified vulnerability and opportunistic-infection categories. The imaging agent characterized the likelihood and anatomical pattern of infection and extracted organism-specific radiological clues. Imaging clues were retained as clinical context but were not treated as direct organism confirmation, which required culture, FilmArray, galactomannan or molecular evidence.

Each organism in the mNGS report was assigned to its source specimen and anatomical site. A rule-based summary then represented, for each organism, the evidence from each modality (support, no support or unknown), whether the specimen was anatomically aligned with the suspected infection, and the relevant host context. Severe host vulnerability broadened consideration of otherwise borderline organisms but could not by itself cause selection (Supplementary Tables 1 and 2).

Characterization of mNGS signals

Each organism in the mNGS report was characterized by its read count, its relative abundance among organisms detected in the same run, its rank against negative controls, its dominance within the specimen and the anatomical relevance of that specimen (Supplementary Table 1). Repeated detections in one patient were resolved by a prespecified hierarchy based on negative-control rank and read burden. Organism names were normalized with a fixed dictionary that merged synonyms (for example, CMV and human cytomegalovirus) but did not collapse distinct species.

Initial pathogen selection

The initial scorer combined the mNGS features with specimen alignment, direct hospital-test support and host context. Each organism was assigned to one of five ordered causative levels, from strongly supported to background (Supplementary Table 2). Strong support required a favourable negative-control rank together with high abundance or within-specimen dominance. Intermediate signals could be strengthened by exact-organism evidence from another microbiological test or by prespecified rules for clinically important pathogens; weak or anatomically discordant signals were downgraded.

Before selection, prespecified guardrails reduced common sources of overinterpretation, including respiratory *Candida* without invasive evidence, oral and aspiration flora, skin and environmental organisms, reactivated or low-specificity viruses and weak non-dominant detections (Supplementary Table 3). Strongly supported organisms were selected directly. Borderline organisms required compatible clinical context, direct microbiological support or a pathogen-specific rule; host vulnerability alone was insufficient. Respiratory *Candida* and generic yeasts always required evidence of invasive infection.

When no organism met the selection criteria, the strongest candidate that passed the exclusion rules was reported as the best available. The system could also consider an organism absent from the mNGS report when an exact-species culture, FilmArray or galactomannan result, or targeted molecular assay provided direct evidence from a clinically relevant specimen. Selection

in this route depended on the hospital test rather than sequencing features. IgM serology alone was not treated as direct evidence; *Chlamydophila* or *Mycoplasma pneumoniae* supported only by IgM was retained for contextual review.

Safety review of unselected organisms

A language-model safety review (GPT-5.6 Luna) examined organisms not selected by the initial rules. It received the deterministic result, the structured hospital evidence and specimen assignments, but could neither remove nor reorder an initial selection. Its prompt emphasized that immunocompromise or membership in a clinically important pathogen group was insufficient for selection on its own and restated the prespecified rules for colonization, reactivation, oral flora, environmental organisms and *Candida* (Supplementary Note 1).

The review separated organisms warranting further evidence assessment from those with low specificity or insufficient support. A rule-based check then standardized organism names and removed candidates supported only by anatomically irrelevant specimens or weak clinical context. Thus, the review could expand the set undergoing evidence assessment but could not promote an organism directly to the final pathogen list.

Literature evidence and patient-level case matching

For every initial selection and additional candidate retained by the safety review, PubMed was searched through NCBI E-utilities using the organism name and lower-respiratory-tract infection terms. Searches included publications available by 30 June 2026 and returned up to ten articles per organism. A language model (GPT-5.6 Luna) assessed each title and abstract for human clinical evidence that the exact species causes infection at the relevant anatomical site. Supporting evidence required an exact-species match, human infection at the target site and a traceable excerpt from the source. When too few articles could be assessed, the literature analysis abstained rather than treating limited evidence as evidence against infection (module A; Supplementary Table 4).

The same articles were then compared with a label-blind summary of the individual patient (module A1). This summary included specimen type and site, clinical syndrome, host factors, imaging and laboratory phenotype and timing, but excluded the reference label, initial selections, read counts, culture and PCR results and all previous review rationales. The model extracted the reported disease context and graded agreement in anatomical site, syndrome, specimen, host, phenotype and timing. Deterministic rules converted these structured observations to strong match, partial match, mismatch or insufficient evidence. Exact-species causal evidence with compatible site and syndrome was required for a positive match; explicit colonization, a conflicting site or syndrome, non-human evidence or broader taxonomic evidence could not support rescue. Only articles with traceable supporting or opposing excerpts entered the aggregate case-match score.

Independent evidence dimensions

Three additional dimensions summarized evidence independent of article case matching (Supplementary Table 4). Sequencing strength captured both absolute burden and relative prominence within the specimen (module B). Direct microbiological evidence was graded by assay specificity and anatomical relevance, with exact-species detection in a sterile site or tissue receiving the strongest grade (module C). Results without a usable specimen site or with contamination concerns were treated as context rather than confirmation, and respiratory *Candida* culture alone was not considered evidence of invasive disease. A final check recorded anatomical coherence and colonization concerns for audit (module D). It did not determine selection, but an explicit anatomical mismatch could not be overridden.

Final evidence integration and pathogen selection

The final rule (R5) retained all organisms selected initially and added a previously unselected organism through either of two rescue routes. The first required exact-species evidence from a sterile site or tissue. The second required a prespecified level of concordance with published cases, anatomical alignment with the suspected infection and either high absolute sequencing burden or strong relative prominence within the specimen. The complete rule and thresholds are reported in Supplementary Table 4.

Initial selections retained their original order, and rescued organisms were appended without an inferred clinical ranking. For each patient, the pipeline recorded the selection route for every organism and the reason each remaining candidate was not selected. Failure to satisfy a rescue route was reported as insufficient evidence for selection, not as evidence that infection was absent.

Explanation generation, audit and delivery

After final selection, candidate-specific microbiology within two days before or after collection of the mNGS specimen was assembled for explanation; the complete time course was retained separately. A language model (GPT-5.6 Luna) generated one explanation for each selected organism from these evidence views and the fixed final decision. The generator was instructed that the selected organisms and their order were immutable. Automated checks required exactly one entry per selected organism, prohibited any organism outside the selected list and verified the presence of all required fields; a failed check blocked delivery.

The delivered record stated the disposition of every organism detected by any method. For selected organisms, it reported the selection route, supporting and opposing findings, relevant literature and sequencing evidence, and specimen site and timing. For non-selected organisms, it reported the applicable exclusion rule or unmet evidence criterion and clarified that non-selection did not exclude infection. Each statement was linked to the corresponding structured evidence. Outputs were delivered as a summary table and an auditable record (Fig. 4) and were labelled as research results pending clinical review.

Conventional read-out comparators

Each conventional read-out was scored at face value by treating its reported organisms as predictions: every organism on the mNGS report; every FilmArray or galactomannan target reported positive; and every organism isolated in culture within the evidence window. [Confirm these definitions and the window against the scoring script, and state the handling of normal flora, no growth and pending results.] No read-out was reinterpreted or filtered. These comparators represent the information a physician receives before any integration.

Direct-prompting comparison

To determine whether performance arose from the structured pipeline rather than from access to a language model alone, four general-purpose models were evaluated by direct prompting in 41 patients. Each model received a single request containing the same nine standardized clinical sections and the laboratory mNGS report, including organism names, read counts, negative-control ranks and specimen types. Direct identifiers and all fields derived from the reference standard or from prior pipeline decisions were removed.

The Traditional Chinese prompt asked the model to act as an infectious-diseases reviewer and emphasized that organism detection does not establish causation. It specified common concerns involving colonization, anatomical site, *Candida*, environmental organisms and viral reactivation; allowed zero or more pathogens; required species-level predictions; and requested supporting evidence, opposing evidence, excluded organisms and limitations in a structured response (Supplementary Note 2). GPT-5.6 Sol, GPT-5.6 Luna, GPT-5.6 Terra and GPT-6 Astra (OpenAI) were queried through the Responses API with medium reasoning effort, a JSON output schema and a maximum of 4,000 output tokens. Tools, external retrieval and conversation history were disabled. Each model was run once per patient, and the response, prompt hash and payload hash were stored before scoring.

Outcome measures and scoring

The unit of evaluation was the (patient, organism) pair. Predicted and reference names were normalized using a fixed synonym table (for example, CMV to human cytomegalovirus and PJP to *P. jirovecii*) and matched at exact-species level. Genus-level matches were not accepted; thus, a prediction of *Candida* against a reference of *Candida tropicalis* contributed one false positive and one false negative.

The primary outcomes were precision, recall and F1, micro-averaged across all patient-organism pairs. Secondary outcomes were the corresponding patient-level macro-averages; the proportions of patients in whom at least one, all or exactly the reference set of pathogens was identified; and the number of patients for whom a method returned no organism (Supplementary Note 5). Both evaluation cohorts required at least one adjudicated pathogen, so precision does not capture false-positive predictions in pathogen-negative patients. Explanations were tested for structural completeness and consistency with the fixed decision, but were not independently rated for clinical usefulness or clause-level faithfulness to the source record; [a reader study rating the delivered explanations, and the effect of the explanation on antimicrobial decisions,

remain to be done].

Secondary registry analyses

The following analyses were exploratory and were not used to develop or evaluate the pathogen-selection rules.

Concordance of sequencing and conventional microbiology

To characterize the information provided by the mNGS report before multimodal integration, we analysed registry cases with an adjudicated pathogen list in the shared workbook ($n = 33$; Supplementary Note 6). KMH cases were not included in this analysis because their adjudications were stored separately in per-patient workbooks. For each case, we compared the organisms and read counts in the mNGS report with organisms detected by culture and non-culture tests and with the adjudicated pathogen list. Names were normalized and matched at exact-species level using the same rules as the primary evaluation. Entries explicitly indicating that no pathogen was present were excluded.

An mNGS detection was considered corroborated when the same species was identified by any conventional test in the same patient. For adjudicated pathogens absent from the mNGS report, we recorded the detecting assay, specimen type and collection time relative to receipt of the mNGS specimen. Conventional assays included culture, PCR and multiplex panels, viral-load testing, antigen testing and IgM or antibody serology. Bronchoalveolar lavage fluid and sputum were classified as non-sterile specimens; blood, cerebrospinal fluid, tissue, ascites, pus and abscess material were classified as sterile. The ability of read count alone to distinguish causative from non-causative detections was summarized by the area under the receiver operating characteristic curve, computed for the 80 of 104 detections with a recorded read count.

Composition and timing of the clinical record

To quantify the evidence available for physician interpretation (Fig. 4a–d), we analysed the complete per-admission source workbooks from KMH. These records contained underlying conditions; admission, transfer and discharge diagnoses; culture reports with collection and reporting times; full-text radiology reports; all syndromic-panel targets; galactomannan measurements; and blood count and chemistry results. Unlike the registry summary, they retained negative results, timestamps and the original report wording.

Of 36 workbooks, 32 distinct admissions were analysed. One lacked microbiology, imaging and panel data; one was labelled as a duplicate; and two were byte-identical copies of another admission that had undergone repeat mNGS. Repeat sequencing of the same admission was counted once. We defined a result as one culture report, radiology study, panel target, galactomannan measurement or recorded analyte value. Organisms were counted once per admission and read-out after synonym normalization; galactomannan was considered to identify *Aspergillus* at an index of ≥ 0.5 in serum or ≥ 1.0 in lavage, and resistance markers were not counted as organisms. Sterile-site cultures comprised blood, ascites, pleural fluid, tissue, cerebrospinal fluid, abscess, deep-wound and intravascular-catheter specimens. Culture turnaround was the interval from specimen collection to the final report.

Radiology wording was classified as qualified when a report used terms such as “probably”, “suspect”, “favour”, “cannot be excluded”, “questionable”, “equivocal”, “may represent” or “suggestive of”. Formulaic recommendations for clinical correlation or differential diagnosis were not considered qualifications. Only aggregate results from these protected source records were exported for analysis.

For each read-out, we summarized the interval between specimen collection and receipt of the mNGS specimen, the interval between admission and mNGS collection and the proportion of patients in whom that read-out identified an organism. Collection intervals within 90 days were summarized by the median, interquartile range and 5th–95th centiles; larger intervals were treated as date-entry errors. Because the registry records conventional tests almost exclusively when an organism was detected (225 of 226 dated culture entries were positive), these distributions describe the timing of organism-identifying evidence, not the complete testing burden. They therefore cannot estimate culture yield or the effect of previous antimicrobial treatment.

The registry recorded arrival and reporting dates for mNGS specimens, but these dates were identical for more than half of the records and were not used to estimate sequencing turnaround. These analyses were descriptive, and no inferential test was applied. Because adjudicators had access to the mNGS report, comparisons with the reference standard share the same non-independence as the primary evaluation. Only de-identified per-detection data and aggregate counts were exported from the protected registry.

Physiological trajectory around the sequencing report

We descriptively examined physiological trajectories around mNGS specimen collection using machine-readable extracts of the complete KMH discharge record. Each dated laboratory value and vital sign retained a link to its source page. Extracts were available for 49 of 51 KMH registry admissions; 48 also had a dated mNGS specimen and recorded discharge status and were included (31 deaths and 17 survivors).

The collection date was defined as day 0. Measurements were grouped into five windows: days –14 to –1, 0–2, 3–7, 8–14 and 15–30. For each patient and window, we first calculated the median value of each marker so that patients with frequent measurements contributed once per interval; group summaries were then calculated from these patient-level medians. Pre-collection and day 3–7 values were compared with a two-sided Wilcoxon signed-rank test among patients with measurements in both windows, without adjustment for multiple comparisons. Fever was summarized as the maximum temperature within each window and defined as a value above 38 °C. Improvement was defined as movement in the clinically favourable direction between the pre-collection and day 3–7 windows; improvement followed by worsening required a subsequent reversal during days 8–30.

Values outside prespecified physiological ranges were excluded. Temperature values were reconstructed from the source table layout and underwent a documented correction for a systematic transcription error, which was checked against adjacent readings from the same admission.

This analysis was observational and descriptive. All patients underwent mNGS, and there was no unexposed comparator. Antipyretics, corticosteroids, sedation and ventilator settings could directly affect temperature and gas exchange but were not available in a form suitable for

adjustment. Antimicrobial start and stop dates were also unavailable; therefore, neither time from reporting to treatment modification nor coverage of the adjudicated pathogen could be determined. The analysis was not used to make causal claims or to propose a monitoring test.

Registry treatment and outcome analysis

Treatment modification, its direction, its attribution to mNGS, mechanical ventilation and discharge status were extracted for all 105 registry specimens and summarized as counts and percentages. Denominators were reported when attribution or ventilation status was missing. The number of organisms per mNGS report was compared across specimen types and patient strata. Because specimen type was associated with disease severity, stratum comparisons were repeated within bronchoalveolar-lavage specimens, and only these restricted comparisons were interpreted. Organism-level agreement with culture was calculated after the same name normalization used in the primary evaluation and cross-checked against the registry’s existing comparison.

Because every registry patient underwent mNGS, the data could not estimate the effect of sequencing relative to no sequencing. We therefore compared patients whose treatment modification was retrospectively attributed by the treating team to mNGS with those whose modification was not. This attribution was neither assigned nor assessed under blinding. In-hospital mortality was compared with Fisher’s exact test. Odds ratios and 95% confidence intervals used the Haldane–Anscombe correction and were additionally estimated with logistic regression adjusted for intensive care at sampling, acute respiratory distress syndrome, active malignancy, corticosteroid treatment and age; Wald intervals are reported.

Comparisons between patients whose treatment changed for any reason and those whose treatment did not were reported descriptively because of evident confounding by indication. The registry did not contain antimicrobial names or administration dates, precluding assessment of treatment appropriateness or time to appropriate therapy. Time-to-event outcomes were also not analysed because patients had to survive long enough to receive a change attributed to a specimen collected after admission. All tests were two-sided, without adjustment for multiple comparisons. These secondary analyses were considered hypothesis-generating, and only aggregate results were exported from the protected registry.

Computational reproducibility

A single orchestrator validated the input to each stage, executed the stages in a fixed order and recorded all commands and output hashes. It contained no pathogen-selection logic and did not overwrite previous outputs. All reported final decisions were regenerated from archived intermediate files, and direct-prompting metrics were recomputed from stored model responses. The current code release reproduced the selected organisms for every patient in the primary cohort. Two implementation differences from the version used for the reported runs did not alter any selection and are described in Supplementary Note 4. The test suite runs offline without network access or credentials.

Statistics and reproducibility

No statistical method was used to predetermine sample size; the cohorts included every registry patient who met the eligibility criteria when the datasets were finalized. Patients were not randomized because all methods were evaluated on the same records. Reference labels were withheld from the language-model stages, and each method’s predictions were recorded before labels were accessed by the evaluation scripts. However, the investigators who developed the rule-based scoring system had access to the labels, and no patients were held out during rule development. The reported diagnostic performance is therefore a development-set result (Discussion).

Patients were excluded only for missing inputs and before model inference (Extended Data Fig. 1). The clinical records were not publicly available and were not used to train the evaluated language models. Precision, recall and F1 are reported as point estimates. Rule-based stages were exactly reproducible, whereas each language-model stage was run once per patient and its output was retained; the reported results therefore do not capture run-to-run model variability. [Add patient-level bootstrap 95% confidence intervals and paired differences in F1 versus each comparator.]

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The patient-level clinical records, mNGS reports, reference labels, model inputs and model outputs contain protected health information and cannot be shared publicly. Aggregate cohort characteristics are given in Table 1. De-identified per-patient evaluation tables (predicted and reference organism sets for AETIA, each conventional read-out and each directly prompted model) will be [provided as Supplementary Data / made available on reasonable request]. PubMed identifiers and stored article judgements from the literature-retrieval and patient-matching analyses will be provided as Supplementary Data. Source data for Figs. 2 and 3 are provided with this paper.

Code availability

The AETIA pipeline, including all rules, prompts, configuration files, synthetic example data and the offline test suite, is available at <https://github.com/OliveWoo/Multimodal-Diagnosis-Model>; the version used for this manuscript is commit d82c0f7. Reproducing the language-model stages requires OpenAI API access and PubMed access; the deterministic stages and the replay of the reported decisions do not. The code will be released under [licence], with the version used here archived at Zenodo ([DOI]). The scripts that generate the figures and tables of this manuscript are available at <https://github.com/ythuang0522/aetia-paper>.

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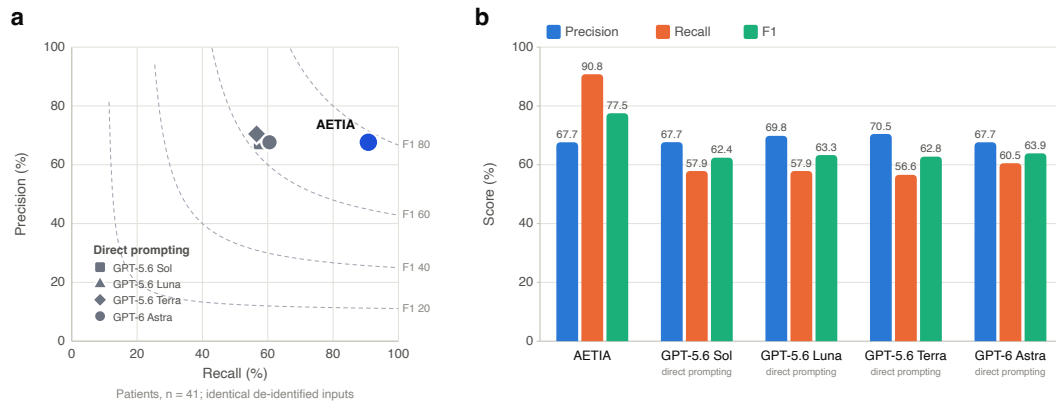


Figure 3: **Comparison with general-purpose language models prompted directly on the same data.** GPT-5.6 Sol, GPT-5.6 Luna, GPT-5.6 Terra and GPT-6 Astra received the same de-identified record, comprising the laboratory mNGS report with per-organism read counts, cultures, FilmArray, galactomannan, molecular tests, imaging reports, blood counts and diagnoses. Each model received one prompt and a JSON schema, without retrieval or reference labels (Methods; Supplementary Note 2). **a**, Precision against recall; AETIA in blue, models in grey (marker key in panel); dashed curves, constant F1. **b**, Precision, recall and F1. $n = 41$ patients (KMUH 30; TVGH and TSGH 11); micro-averaged over (patient, organism) pairs; AETIA values from the frozen pipeline decisions on the same patients (TP 69, FP 33, FN 7).

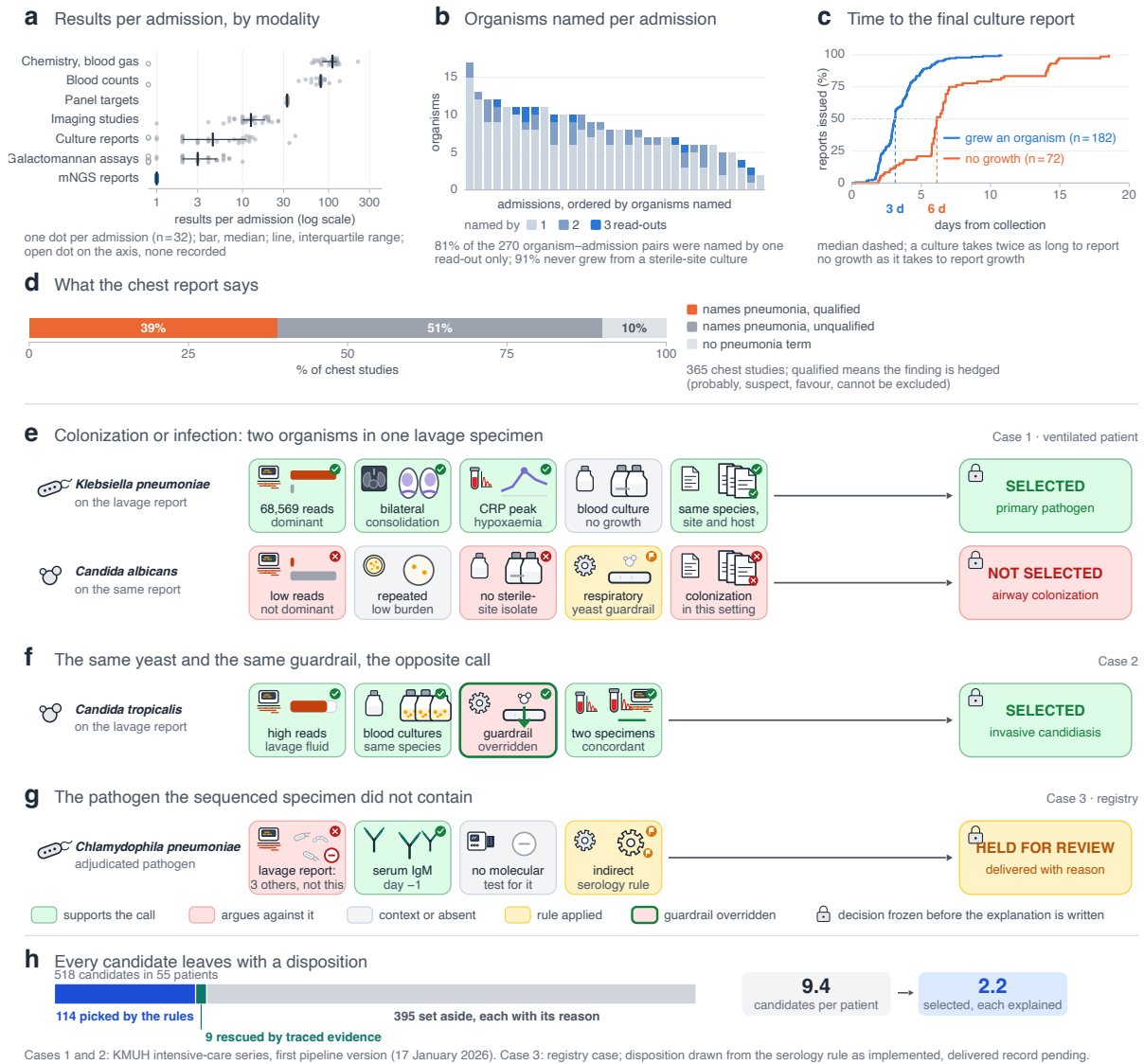
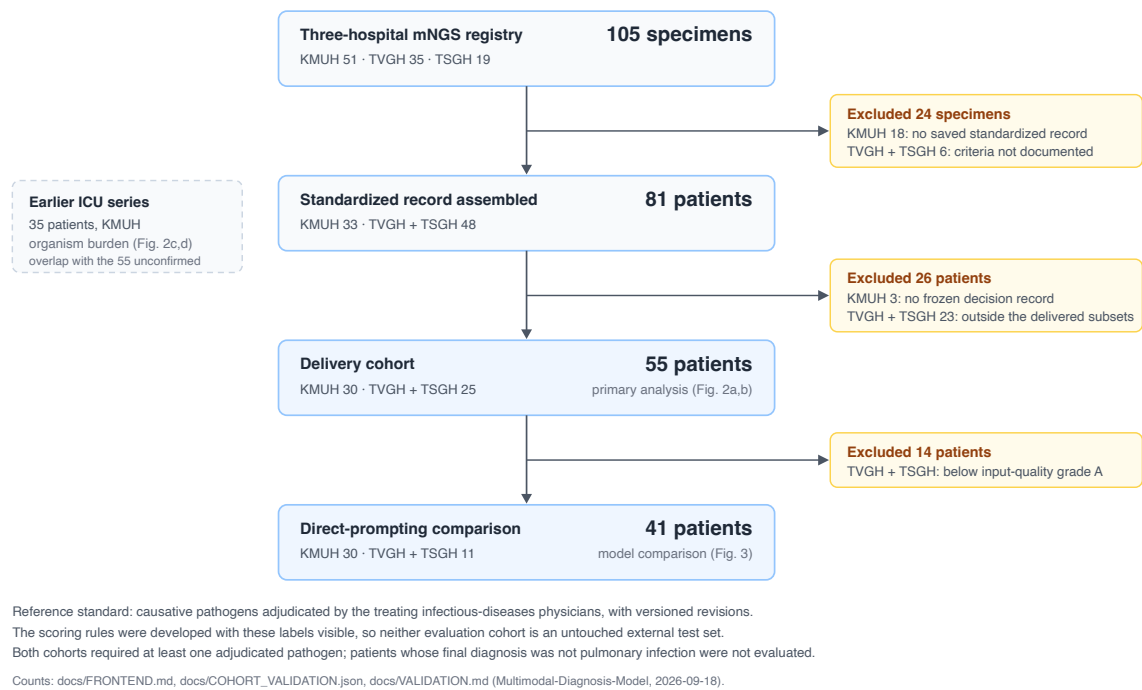
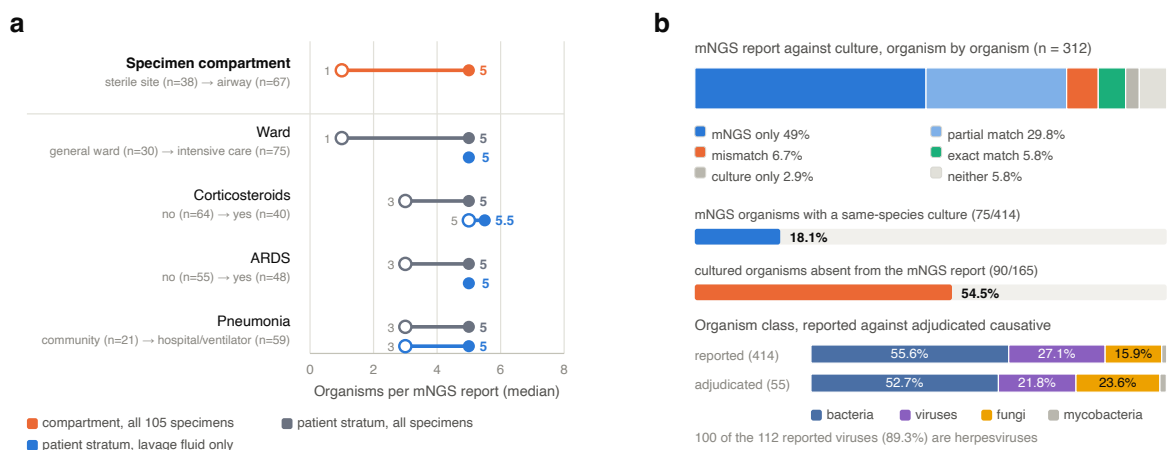


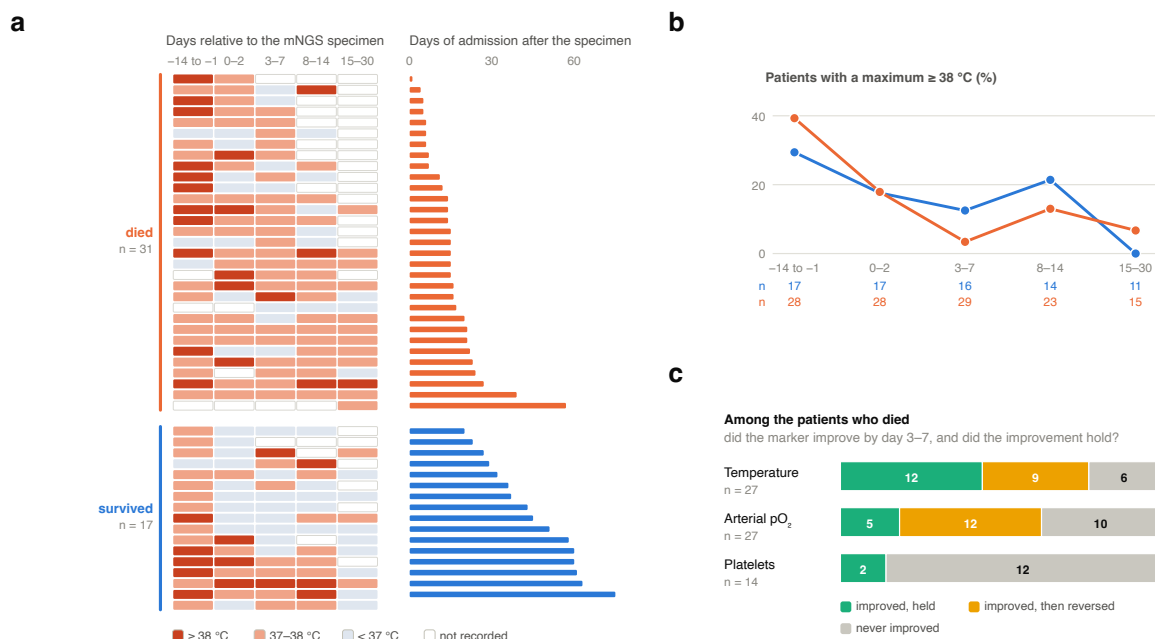
Figure 4: Clinical evidence underlying individual pathogen-selection decisions. **a–d**, Analysis of 32 KMH admissions with available source workbooks; admissions sequenced more than once were counted once. **a**, Number of recorded results per admission by modality, compared with the single mNGS report (median 247, IQR 214–263; new results recorded on 16 of 31 days). **b**, Organisms named per admission (median 8, IQR 6–11). Each bar represents one admission and shading indicates the number of read-outs that named an organism. Of 270 organism-admission pairs, 81% were named by one read-out and 91% were never isolated from a sterile site. **c**, Cumulative percentage of 254 culture reports issued by day and outcome. The median reporting time was 6 days for no growth and 3 days for growth. **d**, Wording of 365 chest-imaging reports: 39% qualified a finding of pneumonia, infiltrate or consolidation, 51% reported such a finding without qualification and 10% used none of these terms. Qualifying terms are defined in Methods. **e–g**, Delivered records for three cases, with one glyph per evidence source and the decision and supporting evidence fixed before explanation generation. **e**, *Candida albicans* classified as airway colonization and *Klebsiella pneumoniae* selected from the same lavage specimen. **f**, *Candida tropicalis* selected because blood cultures grew the same species (overridden guardrail, ringed). **g**, Registry pathogen absent from the sequenced specimen and retained for review on the basis of IgM serology. **h**, Disposition of 518 organisms in 55 patients: 114 selected by the deterministic rules, 9 added by traced evidence and 395 excluded with a recorded reason. [Panels e and f use the two prototype vignettes of the 2026-01-17 talk; regenerate from the delivered pipeline records and obtain case-level approval.]



Extended Data Fig. 1: **Cohort flow.** A single pooled flow from the three-hospital mNGS registry to the two evaluation cohorts, with the contribution of each hospital given at every node and exclusions branching to the right. The two hospital lines differ in where their adjudication is stored and in how their standardized records were assembled, which is a matter of data provenance rather than of eligibility (Methods). Amber, exclusion steps; the criteria for the two-hospital steps are incompletely documented in the source materials [state the 54 → 48 → 25 reasons, and the number of patients excluded because the final diagnosis was not pulmonary infection]. Dashed grey, the earlier 35-patient intensive-care series used for the organism-burden comparison (Fig. 2c,d), which is not part of this flow and whose overlap with the evaluation cohorts is [unconfirmed]. Counts from the repository’s cohort validation of 2026-09-18.



Extended Data Fig. 2: **What the registry’s mNGS reports contain, and how they compare with culture.** The three-hospital registry of 105 mNGS specimens (Table 1). **a**, Organisms per mNGS report (median). Each row is a dumbbell joining the two levels of one contrast; the open circle is the lower level, the filled circle the higher, and the number beside each circle is that level’s median. Orange, the specimen compartment (sterile-site against airway specimens, all 105); grey, a patient stratum over all 105 specimens; blue, the same stratum restricted to bronchoalveolar lavage. The ward, corticosteroid and ARDS gaps present over all specimens close once the compartment is held fixed, so the length of the report tracks the specimen and not the severity of illness; only the pneumonia-category difference persists. **b**, Agreement between the mNGS report and culture. Top, the registry’s own organism-by-organism comparison (312 comparisons; the workbook’s category names are used unchanged, and it does not define partial match or mismatch). Middle, the two corroboration rates, with numerator and denominator given. Bottom, organism class among the 414 reported organisms and the 55 adjudicated causative pathogens. Source: `tables/make_impact.py`.



Extended Data Fig. 3: **The course after the sequencing report, under conventional care.** The 48 KMH admissions with complete machine-readable discharge records (31 deaths, 17 survivors); day 0 is the mNGS specimen. No patient in this analysis received AETIA: these are the trajectories of routine care that motivate the work. **a**, One row per admission, grouped by outcome and ordered by length of admission. Left, the highest temperature recorded in each window after the specimen; right, the days of admission that followed it. The temperature block empties from left to right in both groups: fever resolves whether or not the admission ends in death. **b**, The same temperatures as the proportion of patients reaching 38 °C, with the number contributing to each point. **c**, Counted per patient rather than per window, for the patients who died: whether each marker improved by day 3–7 and whether the improvement held to day 8–30. Temperature and oxygenation improved in most and reversed in about half; platelets, which tracked the outcome, improved in 2 of 14. Windows are –14 to –1, 0–2, 3–7, 8–14 and 15–30 days. Antipyretics, corticosteroids, sedation and ventilator settings act on these variables and are not adjusted for; no causal claim is made (Methods).